Foundations of Point-to-Point Protocol for Network Communication

Arthur Guimarães[1], Marina Schwerz Bon[1], Pedro Webber[1]

[1]Univeridade do Vale do Rio dos Sinos (Unisinos)
Porto Alegre – RS – Brazil

leiteartur@edu.unisinos.br

MarinaBon@edu.unisinos.br

WebberPedro@edu.unisinos.br

Abstract. *This article presents an extensive analysis of the Point-to-Point Protocol (PPP), a fundamental communication protocol in computer networking, employing a theoretical research approach grounded in existing literature. It explores PPP's historical evolution, core mechanisms, security considerations, the intricacies of negotiating network parameters, and a deeper analysis of its functioning. Through the synthesis of current literature, this research offers a deep understanding of PPP's significance in contemporary networking, its flexibility in navigating dynamic landscapes, and its applications in modern technological contexts.*

1. Introduction

The Point-to-Point Protocol (PPP) is a foundational communication protocol that has played a central role in computer networking for several decades. Its robust and adaptable design has made it an essential tool for establishing direct connections between network nodes, enabling data transmission over various physical mediums.

Furthermore, PPP's core mechanisms are multifaceted and integral to its functionality. At its base, PPP secures an establishment and configuration of connections between network nodes. This process involves the negotiation of parameters such as data rates, authentication methods, and error-handling mechanisms. The ability to adapt to various requirements and conditions makes PPP a versatile choice for establishing point-to-point links.

Moreover, Data framing is another fundamental aspect of PPP. It excels in segmenting data into packets, encapsulating them with headers that provide essential addressing and control information. This enables precise routing and efficient data transmission, regardless of the physical medium used.

In addition, PPP's adaptability shines through in its network parameter negotiation. Protocols like the Internet Protocol Control Protocol (IPCP) enable dynamic configuration of crucial parameters, including IP addresses and compression settings. This adaptability allows PPP to excel in various networking environments, from traditional dial-up connections to modern broadband and wireless networks.

In conclusion, In the upcoming sections of this article, we will delve further into the fundamentals of the Point-to-Point Protocol. Building upon the aspects introduced earlier, we will conduct a comprehensive analysis, exploring its historical evolution, core mechanisms, security considerations, and adaptability across various networking environments. The following section will introduce the Point-to-Point Protocol and its fundamental workings and purpose. Following that, a historical overview of the protocol will be provided in section 3. Furthermore, section 4 analyzes the applications and uses of PPP. In contrast, Section 5 and its subsections will explore the protocol's phases. Moreover, Section 6 and its subsections will delve into PPP characteristics and inner workings. Additionally, Section 7 will focus on the protocol's advantages and disadvantages. Section 8 will concentrate on PPP security considerations, while Section 9 will offer a conclusion to this article.

2. Introduction to Point-to-Point Protocol

Firstly, Point-to-Point Protocol is a data link layer (Layer 2) communication protocol used in computer networking. It provides a method for transmitting data packets over serial connections, such as those established over dial-up connections, DSL (Digital Subscriber Line), ISDN (Integrated Services Digital Network), and other technologies. The protocol is primarily used for two key functions:

- Establishing a Link: it is used to establish a connection between two network nodes, often between a user's computer and an internet service provider (ISP). It handles the negotiation and authentication process necessary to create a link between the two devices.
- Encapsulation and Data Transmission: Once the link is established, the protocol encapsulates and transports various types of network-layer packets.

Furthermore, key features of PPP include:

- Error Detection and Correction: it can detect and correct errors in data transmission, ensuring the integrity of the data being transferred.
- Authentication: it supports various authentication methods, allowing a user or device to verify their identity before establishing a connection.
- Compression: the protocol supports data compression, which can increase data transfer efficiency.
- Multiprotocol Support: PPP is capable of transmitting multiple network-layer protocols, making it suitable for transporting different types of data.

In conclusion, the protocol has been widely used for connecting to the internet via dial-up and broadband connections. While newer technologies like DSL and cable have largely replaced dial-up connections for home users, PPP remains a fundamental protocol for establishing and maintaining communication links in various networking scenarios.

3. Historical Development

According to author [Kota1 2015], PPP was initially introduced as an Internet standard in the early 1990s through the collaborative efforts of a working group within the Internet Activities Board's Internet Engineering Task Force (IETF). The primary objective of the group was to establish a standardized method for encapsulating the IP protocol in point-to-point communications, particularly in heterogeneous environments featuring various vendors and devices.

The history and development of PPP reveal that it emerged at a crucial moment in networking. Many of PPP's early advocates were router vendors who encountered challenges when dealing with existing data link protocols, which were predominantly proprietary and vendor-specific. The proliferation of proprietary protocols made achieving internetwork connectivity challenging, especially over serial dial-up or leased lines. In this landscape, the absence of a standardized protocol for high-speed communication added complexity to the network landscape.

Correspondingly, the development and adoption of PPP were pivotal in addressing these challenges. It provided a unifying and open standard for point-to-point communication, allowing for greater interoperability among devices and vendors. The collaborative nature of its development within the IETF ensured that it would serve as a common, non-proprietary foundation for data transmission.

In the present day, millions of Internet users continue to rely on PPP to connect their home computers to their Internet Service Providers for Internet access. Moreover, there remains a substantial portion of individuals who necessitate direct computer-to-computer connections without traversing the internet, often relying on dial-up or leased telephone lines. While the physical connection is established through the telephone line, using an effective point-to-point link protocol, such as PPP, remains essential to manage and control data transfer efficiently. The historical development of PPP stands as a testament to the internet's growth and evolution, providing a foundational protocol that continues to be relevant in modern networking environments.

4. Uses of PPP

As per [Andrew Froehlich 2022], the Point-to-Point Protocol (PPP) stands out as a versatile, bidirectional communication protocol compatible with a diverse array of physical transmission media, including twisted-pair copper wire, fiber optic lines, and satellite links. The adaptability of PPP correlates with its capacity to provide services across a wide spectrum of mediums. These encompass traditional dial-up modems as well as contemporary, highly secure Secure Sockets Layer (SSL)-encrypted virtual private network (VPN) connections. In practical terms, PPP harnesses a modified version of High-level Data Link Control for encapsulating packets, consequently, facilitating robust data transmission.

Consider, for instance, a high-security application within a corporate network. It can forge a connection with the network through a VPN, establishing a secure SSL link. Subsequently, the application's client can initiate the creation of a PPP tunnel, which acts as a conduit for transporting IP packets to the server, ensuring secure and efficient communication.

Moreover, PPP extends its utility to IP tunnelling and other network Layer 3 data between directly connected nodes, independently of whether they are linked by a physical connection or a direct link. Given that IP and TCP inherently lack support for point-to-point connections, PPP emerges as an indispensable solution for enabling such connections across a broad spectrum of physical media, including Ethernet.

In conclusion, point-to-point protocol is considered an integral component of the TCP/IP protocol suite. Notably, variations of PPP have emerged for deployment over Ethernet, exemplified by the PPP over Ethernet (PPPoE) specification. PPPoE, in particular, remains prevalent in contemporary usage, notably among ISPs employing Digital Subscriber Line (DSL) technology. However, this intricate protocol is often concealed from the direct view of end-users.

5. PPP Protocol Phases

The Point-to-Point Protocol (PPP) architecture is a fundamental framework for establishing and maintaining communication links over various physical media, enabling data exchange between network nodes. In the following subsections, the main components of PPP architecture will be introduced [Simpson 1992].

```
+------+        +-----------+           +--------------+
|      | UP     |           | OPENED    |              | SUCCESS/NONE
| Dead |------->| Establish |---------->| Authenticate |--+
|      |        |           |           |              |  |
+------+        +-----------+           +--------------+  |
   ^          FAIL |                   FAIL |              |
   +<--------------+             +----------+              |
   |                             |                         |
   |            +-----------+    |           +---------+   |
   |        DOWN |           |    |   CLOSING |         |  |
   +------------| Terminate |<---+<----------| Network |<-+
                |           |                |         |
                +-----------+                +---------+
```

Figure 1. Visual representation of the point-to-point protocol's phases. [Simpson 1992]

5.1. Link Establishment

This phase serves as both the beginning and conclusion of the link. When external triggers, such as carrier detection or network administrator configurations, indicate the readiness of the physical layer for operation, PPP seamlessly progresses into the Link Establishment phase. Within this phase, the LCP automaton, as detailed subsequently, operates in either the Initial or Starting states. The transition to the Link Establishment phase initiates an Up event within the automaton, marking the formal initiation of the link.

5.2. Link Control Protocol (LCP)

The Link Control Protocol (LCP) assumes a pivotal role in the establishment, configuration, maintenance, and termination of point-to-point connections. It orchestrates these operations through four distinct phases.

In the first phase, known as Link Establishment and Configuration Negotiation (Phase 1), LCP initiates the connection by exchanging Configure packets, allowing the exchange of network-layer datagrams like IP packets. The transition to the Open state occurs when both Configure-Ack packets have been successfully sent and received. It's noteworthy that LCP exclusively handles link configuration, not the configuration of specific network-layer protocols. Parameters unrelated to particular protocols are configured by LCP, with Configuration Options set to default values unless modified during the exchange.

Following this, in Phase 2, the optional Link Quality Determination phase takes place. Here, the link's quality is evaluated to ensure it's suitable for activating network-layer protocols. This phase is discretionary, and LCP may delay the transmission of network-layer protocol data until its completion. The specific method for Link Quality Determination can vary across implementations, as long as it adheres to LCP guidelines. One suggested approach involves employing LCP Echo-Request and Echo-Reply packets. It's vital to note that this phase's duration can vary, and therefore, fixed timeouts should be avoided while waiting for progression to Phase 3.

Once the Link Quality Determination phase concludes, in Phase 3, Network Control Protocols (NCP) can individually configure network-layer protocols, enabling their activation or deactivation as needed. If LCP initiates a link termination, it communicates this to the network-layer protocols, allowing them to respond appropriately.

Lastly, in Phase 4, LCP retains the capability to terminate the link at any time. Typically, this action is initiated by a human user, but it can also occur due to specific physical events like carrier loss or the expiration of an idle-period timer.

Furthermore, LCP operates under the governance of defined packet formats and finite-state automation. LCP packets fall into three distinct classes:

- Link Establishment Packets: These packets initiate and configure a connection and include Configure-Request, Configure-Ack, Configure-Nak, and Configure-Reject packets.
- Link Termination Packets: These are responsible for concluding a connection and consist of Terminate-Request and Terminate-Ack packets.
- Link Maintenance Packets: Designed for connection management and diagnostics, they encompass Code-Reject, Protocol-Reject, Echo-Request, Echo-Reply, and Discard-Request packets.

Lastly, it's essential to acknowledge that while the protocol takes measures to prevent Configuration Option negotiation loops, it cannot guarantee their complete avoidance. Conflicting policies or slow convergence may still occur. To address this, implementors should consider incorporating loop detection mechanisms or higher-level timeouts. If timeouts are implemented, they should be user-configurable. For example, to mitigate Configure-Request or Terminate-Request livelocks, implementations may employ a maximum retries counter, which determines the number of packet retransmissions before considering a livelock situation. A configurable maximum number of retries is recommended, with a default setting typically at ten (10) retransmissions.

5.2.1. LCP Events

Firstly, actions and transitions within the LCP state machine result from events. Furthermore, several events stem from commands executed locally, such as Active-Open, Passive-Open, and Close. While others are triggered by the reception of packets from the remote end. Which include Receive-Configure-Request, Receive-Configure-Ack, Receive-Configure-Nak, Receive-Terminate-Request, and Receive-Terminate-Ack. Finally, certain events are initiated by the expiration of the Restart timer, which is set in motion as a consequence of preceding events, like Timeout. Moreover, LCP events include:

- Active-Open (AO): The occurrence of the Active-Open event signifies the local execution of an Active-Open command by the network administrator, be it a human or a program. When this event takes place, LCP is tasked with promptly initiating a connection by engaging in the exchange of configuration packets with the LCP peer.
- Passive-Open (PO): Similar to the Active-Open event, the Passive-Open event denotes local execution, but with a distinct approach. Instead of an immediate exchange of configuration packets, LCP remains in a waiting state for the peer to dispatch the first packet. This initiation occurs only after an Active-Open event in the LCP peer.
- Close (C): The Close event signals the local execution of a Close command. Upon this event, LCP should immediately initiate the closure of the connection.
- Timeout (TO): The Timeout event indicates the expiration of the LCP Restart timer. This timer is employed to manage the timeouts for Configure-Request and Terminate-Request packets. When the timer expires, a Timeout event is triggered, prompting the retransmission of the corresponding Configure-Request or Terminate-Request packet. The Restart timer is configurable, defaulting to three seconds.
- Receive-Configure-Request (RCR): The Receive-Configure-Request event takes place when a Configure-Request packet is received from the LCP peer. This packet signals the intent to establish an LCP connection and may specify Configuration Options.
- Receive-Configure-Ack (RCA): The Receive-Configure-Ack event begins upon receiving a valid Configure-Ack packet from the LCP peer. This packet serves as an acknowledgement to a Configure-Request packet.
- Receive-Configure-Nak (RCN): The Receive-Configure-Nak event occurs upon receiving a valid Configure-Nak or Configure-Reject packet from the LCP peer, representing a negative response to a Configure-Request packet.
- Receive-Terminate-Request (RTR): The Receive-Terminate-Request event occurs when a Terminate-Request packet is received from the LCP peer, indicating the desire to close the LCP connection.
- Receive-Terminate-Ack (RTA): The Receive-Terminate-Ack event takes place upon receiving a Terminate-Ack packet from the LCP peer, serving as a response to a Terminate-Request packet.
- Receive-Code-Reject (RCJ): The Receive-Code-Reject event unfolds upon receiving a Code-Reject packet from the LCP peer, signalling an error that leads to an immediate connection closure.
- Receive-Unknown-Code (RUC): The Receive-Unknown-Code event transpires when an uninterpretable packet is received from the LCP peer, representing a response to an unknown packet.
- Receive-Echo-Request (RER): The Receive-Echo-Request event occurs upon receiving an Echo-Request, Echo-Reply, or Discard-Request packet from the LCP peer. An Echo-Reply packet serves as a response to an Echo-Request packet, with no reply to a Discard-Request.
- Physical-Layer-Down (PLD): The Physical-Layer-Down event takes place when the Physical Layer indicates that it is down.

5.2.2. LCP Actions

Events within the LCP state machine prompt various actions, commonly entailing the transmission of packets and the activation or deactivation of the Restart timer. Additionally, the array of LCP actions encompasses:

- Send-Configure-Request (scr): This action involves the transmission of a Configure-Request packet, signaling the intention to establish an LCP connection with a specified set of Configuration Options. The Restart timer initiates after transmitting the Configure-Request packet to safeguard against potential packet loss.
- Send-Configure-Ack (sca): The Send-Configure-Ack action transmits a Configure-Ack packet, acknowledging the receipt of a Configure-Request packet with an acceptable set of Configuration Options.
- Send-Configure-Nak (scn): The Send-Configure-Nak action transmits a Configure-Nak or Configure-Reject packet, as appropriate, as a negative response to a Configure-Request packet with an unacceptable set of Configuration Options. Configure-Nak packets decline a Configuration Option value and propose a new, acceptable value, while Configure-Reject packets refuse negotiation on a Configuration Option, often due to lack of recognition or implementation.
- Send-Terminate-Req (str): This action entails the transmission of a Terminate-Request packet, indicating the desire to close an LCP connection. The Restart timer is initiated after sending the Terminate-Request packet to mitigate the impact of potential packet loss.
- Send-Terminate-Ack (sta): The Send-Terminate-Request action transmits a Terminate-Ack packet, confirming the receipt of a Terminate-Request packet or validating the belief that an LCP connection is closed.
- Send-Code-Reject (scj): The Send-Code-Reject action transmits a Code-Reject packet, signifying the reception of an unknown packet type. This unrecoverable error prompts an immediate transition to the Closed state on both ends of the link.
- Send-Echo-Reply (ser): The Send-Echo-Reply action involves transmitting an Echo-Reply packet, acknowledging the receipt of an Echo-Request packet.

5.2.3. LCP States

According to [Perkins 1990], there are various states regarding LCP which guarantee its functionality, the following are:

- Closed (1): In the initial and concluding state, the connection remains inactive, rejecting all incoming peer requests. Transitions occur through Active-Open or Passive-Open events. The Restart timer is dormant, allowing the Physical Layer connection to be terminated at any point.
- Listen (2): Resembling the Closed state, the connection is inactive, but incoming peer requests are accepted. Upon receiving a Configure-Request, an immediate response is issued, and the Restart timer begins. Depending on the response, the state transitions to Ack-Sent or Request-Sent. Terminate-Ack packets are dispatched in response to Configure-Ack or Configure-Nak packets.

- Request-Sent (3): This state signifies an active attempt to establish a connection. A Configure-Request is transmitted, and the Restart timer is active. Configuration options are adjusted in response to Configure-Nak or Configure-Reject. The state transitions to Ack-Received upon receiving Configure-Ack or back to Request-Sent when the Restart timer expires.
- Ack-Received (4): Having sent a Configure-Request and received a Configure-Ack, the Restart timer remains active. Upon receiving a Configure-Request with acceptable options, a Configure-Ack is transmitted, and the connection moves to the Open state. It reverts to Request-Sent after the Restart timer expires.
- Ack-Sent (5): In this state, both Configure-Ack and Configure-Request have been sent, awaiting a Configure-Ack. Upon reception, the Restart timer halts, and the state transitions to Open. Configuration options are adjusted and retransmitted upon Configure-Nak or Configure-Reject. The Restart timer restarts when it expires.
- Open (6): A functional connection is established, allowing data transmission. The transition occurs to the Closing state upon receiving a Close command or Terminate-Request. Echo-Requests are met with Echo-Replies, and Terminate-Requests lead to Terminate-Ack transmissions.
- Closing (7): Actively attempting to close the connection, a Terminate-Request is sent, and the Restart timer is active. Transition to the Closed state happens upon receiving Terminate-Ack or after the Restart timer expires. The Passive-Open event is handled cautiously, ensuring a smooth transition to the Closed state.

5.3. Authentication Protocol (AP)

The Authentication Protocol (AP) stands as a fundamental element within the Point-to-Point Protocol (PPP), crafted to elevate the security and dependability of point-to-point connections in the domain of network communication. Its primary objective is to furnish a secure mechanism for validating the identities of devices or users at each terminus of the PPP link before initiating any data exchange. This authentication process is pivotal, serving to thwart unauthorized access and uphold the integrity of data transmissions.

PPP provides a variety of authentication methods, including the Password Authentication Protocol (PAP) and the Challenge Handshake Authentication Protocol (CHAP). PAP, a direct approach, involves the exchange of username and password in plain text, rendering it less secure but more straightforward to implement. In contrast, CHAP employs a more robust mechanism where a challenge is presented, and the responding device uses a confidential key to generate a unique response. This iterative challenge-response process is recurrent to consistently secure the connection.

Functioning as a critical security layer within PPP, the Authentication Protocol exhibits flexibility and adaptability, tailoring its capabilities to meet diverse security requisites in various network environments. Its pivotal role revolves around fortifying the integrity and legitimacy of point-to-point connections, establishing it as an indispensable constituent in the realm of secure network communications.

5.4. Network Control Protocol (NCP)

As per [Perkins 1989], the Network Control Protocol (NCP) is an essential element of the Point-to-Point Protocol (PPP) architecture responsible for the configuration and management of specific network-layer protocols, such as IP, IPX, or AppleTalk. NCPs are designed to ensure that each network-layer protocol operates efficiently over the PPP link by configuring parameters and settings relevant to that protocol.

NCPs work by engaging in a negotiation process with the peer at the other end of the PPP link. This negotiation involves the exchange of control packets that define the specific settings for the network-layer protocol. For instance, when configuring the Internet Protocol (IP), the IP Control Protocol (IPCP) handles negotiations related to IP addresses, DNS server information, and other IP-specific parameters.

The NCP negotiation process aims to establish a common set of parameters that both ends of the PPP link will adhere to, ensuring compatibility and effective data exchange. NCPs can be independently opened or closed at any time, allowing for flexibility in the use and management of various network-layer protocols over the same PPP link. This approach enables seamless interoperability and dynamic adaptation to network requirements, making PPP a versatile and powerful protocol for point-to-point communication.

5.5. Link Termination

An important exception to the earlier mentioned process exists, primarily related to the availability of the LCP Protocol-Reject mechanism. Specifically, when the Link Control Protocol (LCP) is in the Open state, any incoming protocol packet that is incompatible with the implementation is mandated to be returned in the form of a Protocol-Reject packet. In contrast, only those network-layer protocols that are supported by the implementation are processed, while any unsupported protocols are discarded.

This mechanism, the LCP Protocol-Reject, serves as a critical component of the Point-to-Point Protocol (PPP) termination phase. When faced with unsupported protocols, it not only facilitates the rejection of incompatible data but also ensures that the network remains secure and operational. By employing this approach, PPP maintains a robust framework for terminating connections gracefully and securely, while also providing a mechanism for handling exceptions with precision and clarity [Perkins 1989].

6. PPP Protocol Characteristics and Inner-Workings

The upcoming sections will delve into a thorough examination of the characteristics, associated technologies, and intricacies of the Point-to-Point Protocol (PPP). In Section 6.1, the protocol's approach to loop avoidance, its mechanisms and strategies to ensure stable and efficient networking will be mentioned. Moving forward to Section 6.2, there will be a focus on Protocol Encapsulation. Lastly, in Section 6.3, PPP Frame Format will be dissected, and structural elements and specifications that define how information is packaged and exchanged within the protocol are included.

6.1. Loop Avoidance

Loop avoidance is a crucial aspect of network protocols to ensure the stable and efficient operation of data transmission. In the context of the Point-to-Point Protocol (PPP), which is commonly used for establishing direct connections between network devices, loop avoidance mechanisms play a significant role in preventing the formation of loops that could lead to network instability and degraded performance [Perkins 1990].

One primary method employed for loop avoidance in PPP is the implementation of the Spanning Tree Protocol (STP). STP is a standardized protocol that operates at the data link layer and is specifically designed to detect and eliminate loops in network topologies. When PPP is used in conjunction with Ethernet or other LAN technologies, STP helps prevent the formation of redundant paths that could result in loops.

STP accomplishes loop avoidance by designating a single ”root bridge” within the network, from which all paths are considered and optimized. Redundant paths are identified and blocked to create a loop-free topology. In PPP environments, this means that if PPP is running over Ethernet, STP ensures that only one path at a time is active for data transmission, preventing loops and associated issues like broadcast storms and network congestion.

Additionally, PPP itself includes a feature called the Maximum Receive Unit (MRU) to help avoid loops. The MRU defines the maximum size of a PPP frame that a link can support, preventing frames from endlessly circulating in the network and causing loops. In summary, loop avoidance in PPP involves the integration of protocols like STP and the utilization of features such as MRU to prevent the formation of loops in the underlying network infrastructure. This ensures the stability and efficiency of data transmission in PPP-based connections, contributing to the overall reliability of network communication.

6.2. Protocol Encapsulation

A concise overview of the Point-to-Point Protocol (PPP) encapsulation is presented below in Figure 2.

```
+----------+-------------+---------+
| Protocol | Information | Padding |
| 16 bits  |      *      |    *    |
+----------+-------------+---------+
```

Figure 2. Visual representation of PPP encapsulation. [Simpson 1993]

The Protocol field, comprising two octets, serves as the identifier for the datagram enclosed within the Information field of the packet. Notably, this field follows a specific transmission and reception order, with the most significant octet appearing first. It’s crucial to underscore that all protocols intended for use within this field must have an odd numerical value, and the least significant bit of the least significant octet should be set to ”1.” Additionally, all protocols must be assigned in a manner where the least significant bit of the most significant octet is ”0.” Frames that do not adhere to these specified criteria are treated as having an unrecognized Protocol.

Turning to the Information field, it encompasses a variable number of octets and functions as the container for the datagram associated with the protocol identified in the Protocol field. The Information field has a defined maximum length, which includes any necessary padding, referred to as the Maximum Receive Unit (MRU), typically set at 1500 octets by default. PPP implementations, through a negotiation process, possess the flexibility to employ alternative MRU values. During transmission, the Information field may be padded with any number of octets up to the MRU. It’s essential that each protocol can effectively differentiate between padding octets and authentic data, ensuring the integrity of the data encapsulation and transmission process [Simpson 1993].

6.3. Protocol Frame Format

A summary of the PPP encapsulation and frame format is shown below in Figure 3.

```
+----------+---------+---------+----------+------------
|   Flag   | Address | Control | Protocol | Information
| 01111110 | 1111111 | 0000011 | 16 bits  |      *
+----------+---------+---------+----------+------------
        ---+---------+----------+
           |   FCS   |   Flag   |
           | 16 bits | 01111110 |
        ---+---------+----------+
```

Figure 3. Visual representation of PPP frame. [Perkins 1990]

According to [Perkins 1990], the frame structure for standard PPP communication is outlined as well. The fields are transmitted from left to right, adhering to a common Internet practice that orders binary numbers from the Most Significant Bit to the Least Significant Bit, reading from left to right. It's important to note that this differs from standard ISO and CCITT practice, which orders bits as transmitted (network bit order).

The frame structure includes the following elements:

- Flag Sequence: This single octet denotes the start or end of a frame.
- Address Field: A single octet containing the binary sequence 11111111, signifying the All-Stations address. PPP does not assign individual station addresses.
- Control Field: A single octet with the binary sequence 00000011, indicating the Unnumbered Information (UI) command with the final bit set to zero.
- Protocol Field: This field is made up of two octets and identifies the protocol encapsulated in the Information field.
- Information Field: This part of the frame contains the actual datagram for the specified protocol. The end of the Information field is determined by the closing Flag Sequence, with room for the Frame Check Sequence field.
- Frame Check Sequence (FCS) Field: Normally 16 bits, this field can be extended to 32 bits by agreement between PPP implementations for enhanced error detection. The FCS is calculated over specific fields, excluding start and stop bits or additional bits and octets added for transparency.

7. Advantages and Disadvantages of PPP Protocol

The subsequent sections will comprehensively examine the merits and drawbacks associated with the Point-to-Point Protocol (PPP). This analysis will delve into both the favourable aspects that contribute to PPP's effectiveness in networking scenarios and the challenges or limitations that may impact its application and performance.

7.1. Advantages

The Point-to-Point Protocol (PPP) has gained widespread adoption and sustained relevance in networking due to several key advantages. A primary strength of PPP lies in its adaptability, as it can be utilized across various physical transmission media, ranging from traditional copper wire connections to high-speed fiber optic lines and satellite links. This flexibility enables PPP to address a diverse range of networking requirements, spanning from legacy dial-up connections to contemporary and secure Virtual Private Network (VPN) setups, making it suitable for various applications.

One notable feature of PPP is its robust support for authentication mechanisms, including the Password Authentication Protocol (PAP) and the Challenge Handshake Authentication Protocol (CHAP). This capability significantly enhances network security by validating the identity of connecting devices, making PPP an ideal choice for secure connections, such as those established by Internet Service Providers (ISPs) and businesses.

PPP's adaptability extends to its compatibility with a broad array of network-layer protocols, encompassing Internet Protocol (IP), Internetwork Packet Exchange (IPX), and AppleTalk. This versatility positions PPP as a universal data link layer protocol, seamlessly integrating with diverse network architectures and facilitating communication between devices relying on different network-layer protocols.

Furthermore, PPP incorporates error detection and correction mechanisms, ensuring the integrity of data transmission. This feature is particularly crucial for maintaining data accuracy in unreliable or noisy communication channels.

In summary, PPP's advantages include its flexibility, adherence to industry standards, support for authentication, compatibility with multiple network-layer protocols, and error detection and correction capabilities. These qualities collectively establish PPP as a versatile and reliable choice for point-to-point and remote network connections, meeting the demands of modern networking while accommodating legacy systems [Lithuania 2023].

7.2. Disadvantages

Point-to-point Protocol has many characteristics essential for a functioning network, however, it also has a set of disadvantages. One of the notable disadvantages of PPP is its limited support for multicast and broadcast traffic. Designed primarily for point-to-point communication, PPP does not efficiently handle scenarios that involve multicast or broadcast data transmission. This can be a drawback in applications like video streaming, online gaming, or broadcasting, where the ability to transmit data to multiple recipients simultaneously is essential.

Another disadvantage of PPP is its lack of native encryption. While it provides a basic framework for data link layer communication, it does not offer built-in encryption mechanisms. To secure PPP connections, additional protocols like PPTP (Point-to-Point Tunneling Protocol) or L2TP (Layer 2 Tunneling Protocol) are often employed. However, implementing these additional security layers can add complexity to network configurations and may introduce vulnerabilities if not managed correctly.

Furthermore, PPP's error detection and correction capabilities are limited. In environments with high levels of noise or unreliable physical media, PPP may not adequately ensure data integrity. More advanced data link layer protocols offer more robust error handling and correction mechanisms, making them a better choice for such scenarios.

PPP was developed in an era dominated by dial-up connections. This legacy can lead to incompatibility with modern network technologies. Integrating PPP with contemporary network environments such as Ethernet, Wi-Fi, or cellular networks can be challenging and may require additional adaptations or conversion mechanisms. Moreover, PPP has limited routing capabilities, which can be a disadvantage in scenarios requiring complex routing configurations. Advanced routing options may not be readily available with PPP, making it less suitable for networks with sophisticated routing needs.

In summary, while PPP is a reliable and widely used data link layer protocol, its limitations in handling multicast and broadcast traffic, providing native encryption, dealing with complex authentication, error correction, and compatibility with modern network technologies, along with its complexity in configuration and limited routing capabilities, should be considered when evaluating its suitability for specific network implementations. Network administrators must weigh these disadvantages against their specific network requirements and constraints when deciding on the adoption of PPP [Lithuania 2023].

8. Security Considerations of PPP

According to [Kastenholz 1993], PPP offers the advantage of dedicated connections, where each user consistently utilizes the same IP address. However, this can open the door to IP spoofing. Fortunately, PPP comes with robust authentication capabilities designed to thwart IP spoofing attempts. Firstly, you can configure connection profiles comprising a username and an associated password, which relies on validation lists for authentication. Additionally, PPP eliminates the need for a connection script.

Another noteworthy feature of PPP is the option to implement the Challenge Handshake Authentication Protocol (CHAP). With CHAP, concerns about eavesdropping on passwords become obsolete, as CHAP encrypts both usernames and passwords. CHAP is utilized in PPP connections if both endpoints support it. During the signal exchange for establishing communication between two modems, the systems negotiate the use of CHAP. For instance, if system A supports CHAP while system B does not, system A can choose to either deny the session or agree to use unencrypted user credentials. This agreement to use unencrypted credentials is referred to as ”negotiating down.”

The choice to negotiate down is a configurable option. In a secure intranet setting, where you can ensure that all systems support CHAP, it’s advisable to configure the connection profile not to negotiate down. Conversely, on a public connection, where your system initiates the call, you might opt to negotiate down. PPP connection profiles offer the flexibility to specify valid IP addresses. For instance, you can define a specific address or a range of addresses for a particular user, enhancing security against spoofing attempts. To further guard against spoofing or unauthorized access during an active session, PPP can be configured to periodically re-challenge the connection. This means that, at regular intervals, your IBM i platform might re-request user credentials while the PPP session remains active, ensuring it’s the same connection profile in use.

The end-users remain oblivious to this re-challenge process as it transpires beneath their interaction level. PPP also accommodates scenarios where remote LANs dial into your IBM i platform and extended network. In such an environment, enabling IP forwarding is often necessary. However, PPP offers robust safeguards, including password encryption and IP address validation, making it more resistant to potential intruders attempting to establish a network connection.

9. Conclusion

In summary, this theoretical research scientific article has offered a comprehensive analysis of the Point-to-Point Protocol (PPP), unveiling its foundational principles, historical evolution, and contemporary relevance within communication networks. Through a detailed examination of the protocol's architectural intricacies, encapsulation methods, and authentication mechanisms, a deep understanding of its pivotal role in enabling secure and dependable data exchange between two endpoints has been achieved.

In conclusion, in an era characterized by increasingly intricate and interconnected networks, the insights derived from this research serve as invaluable guidance for network engineers, researchers, and policymakers. It is evident that PPP remains an indispensable component of contemporary data transmission, and its ongoing development and integration into emerging technologies play a pivotal role in shaping the trajectory of global communications.

References

Andrew Froehlich, J. B. (2022). Point-to-point protocol (ppp).

Baeldung (2023). Point-to-point protocol (ppp).

Bagci, T. (2023). What is ppp (point-to-point protocol)? — how does it work?

IBM (2021). Security considerations for using point-to-point protocol.

Kastenholz, F. (1993). The definitions of managed objects for the security protocols of the point-to-point protocol.

Kota1, V. (2015). Point to point protocol(ppp).

Kurose, J. (2016). *Computer Networking: A Top-down Approa*. Addison-Wesley, 7th edition.

Lithuania, G. R. (2023). Ppp advantages and disadvantages.

Perkins, D. (1989). The point-to-point protocol: A proposal for multi-protocol transmission of datagrams over point-to-point links.

Perkins, D. (1990). The point-to-point protocol for the transmission of multi-protocol datagrams over point-to-point links.

Point, J. T. (2021). Ppp protocol.

Shahid, K. (2023). What is point-to-point protocol?

Simpson, W. (1992). The point-to-point protocol (ppp) for the transmission of multi-protocol datagrams over point-to-point links.

Simpson, W. (1993). The point-to-point protocol (ppp).

Tanenbaum, A. S. (2022). *Computer Networks*. Pearson Education, 8th edition.